

This three branches produce three different types of features of same dimensional tensor and Here CGAF adaptively weight their contribution. Standard fusion techniques like **Concatenation** or **Cross-Attention** operate on a flawed assumption: that every deepfake leaves the same kind of 'fingerprint.' But in reality, forgery traces are domain-specific. **NeuralTextures** artifacts are strongest in the **Spatial** domain; **Deepfakes** and **FaceShifter** are revealed by **Frequency** artifacts; and **FaceSwap** leaves **Noise** inconsistencies at blending boundaries.

A static weighted sum is suboptimal because the importance of these domains varies per sample. We needed a system that doesn't just 'add' data, but 'interrogates' it."

[Key Pillar 1: The Consistency Check] "Our first stage is the **Consistency Check**. Before we fuse anything, we project our features and calculate **Cosine Similarity Maps**. The logic is simple: we look for agreement. If both the Spatial and Frequency branches detect an anomaly in the same region, the model gains mathematical confidence that a forgery is actually present."

[Key Pillar 2: Gated Filter & Fallback] "Second, we use **Temperature-Scaled Sigmoid Gating**. Unlike standard attention, we use **Log-Parameterization** to keep our temperature bounded between **0.1 and 10.0**. This prevents 'dead gradients' and ensures the model keeps learning.

Crucially, we've implemented a **Conservative Fallback**. We use **Cross-Consistency Modulation**: if the Frequency and Noise branches disagree—represented by a low C_{fn} —the model suppresses those gates. It falls back to the Spatial branch as a 'safe' baseline, preventing 'garbage' signals from ruining the final prediction."

[Key Pillar 3: Global Semantic Oversight] "Finally, we have the 'brain' of the module: **Global Semantic Oversight**. We use high-level features—**F_high**—to generate a global gating signal.

- If the scene looks stable and natural, this signal suppresses the noise and frequency gates to **reduce false positives**.
- If the global context indicates manipulation, it **amplifies** those auxiliary branches to catch even the most subtle artifacts.

Overall CGAF give better result than traditional modules because it doesn't just "add" these domains; it **interrogates** them using a multi-step gated logic

Temporal Modeling: Catching the "Flicker"

[Transition] "Once our **CGAF** module has fused the spatial, frequency, and noise data for each frame, we are left with a rich feature set. But deepfakes aren't just static

images—they are videos. To catch them, we must look at how these features change over time."

[1. Masked Spatial Pooling] "Instead of using basic averaging, which can be 'polluted' by background pixels, we use **Energy-Weighted Masked Pooling**.

- **The Logic:** We calculate the feature energy for every pixel.
- **The Result:** This acts like a soft attention mask, forcing the model to focus on high-activity areas—like the face—while suppressing irrelevant background noise."

[2. Multi-Scale Temporal Differences] "This is where we address a major weakness of single-frame detectors. Video deepfakes often exhibit **temporal flickering** and motion inconsistencies that are invisible in a still photo. We compute frame differences at three distinct temporal scales:

- **Scale 1 (Δ^1):** Captures high-speed jitter and fast GAN inconsistencies.
- **Scale 2 (Δ^2):** Detects short-range temporal drift.
- **Scale 4 (Δ^4):** Identifies medium-range motion errors. By analyzing these three scales, we ensure no artifact—whether it's a quick flash or a slow drift—goes unnoticed."

[3. Sequence Construction & Transformer Encoding] "All of these features—the original frame data and our multi-scale differences—are concatenated into a master sequence.

- **Motion-Aware Bias:** We add a specific 'motion-aware bias' to every token to tell the model which frames contain the most movement.
- **The Transformer:** This sequence is fed into a **2-layer Transformer Encoder**. This is the final decision-maker. It uses multi-head attention to compare every frame and every difference across the entire video, producing a single, high-dimensional 'video-level feature' that summarizes the forgery evidence."

[Final Conclusion for Speech] "By combining **CGAF's** multi-domain analysis with this **Multi-Scale Temporal Difference** encoding, we've built a model that is sensitive to both the microscopic artifacts of AI generation and the macroscopic inconsistencies of video motion."

The final stage of the **AFAGNet v3** architecture moves from feature extraction to the actual decision-making and visual explanation. This is handled by two specialized heads that provide both a final verdict and a precise map of the forgery.

Classification and Localization: The Final Verdict

[1. Classification Head: Context-Aware Decision Making] "The classification head is responsible for the final 'Real or Fake' decision. It doesn't just look at the temporal features (v); it also integrates high-level semantic context (v_{high}).

- **Content-Aware Gating:** We use a learnable gate to determine exactly how much the semantic context should influence the final result.
- **Progressive Dropout:** To ensure the model generalizes well and doesn't overfit to specific datasets, we use a 4-layer MLP with progressive dropout—starting at **0.3** and reaching **0.4** in the middle layers.
- **Numerical Stability:** The head returns raw logits during training for better numerical stability, while the final sigmoid is applied only during inference."

[2. Localization Head: Mapping the Forgery] "One of the strongest features of our model is that it doesn't just say 'this is a fake'—it shows you where. The localization head decodes the fused features back into a 224×224 forgery probability mask.

- **Semantic Skip Connections:** To keep the spatial details sharp, we include a skip connection from F_{high} . This ensures that as we upsample the features, we don't lose the high-level semantic context of the face.
- **Progressive Upsampling:** We use a cascade of **Transposed Convolutions** to step up the resolution from 14×14 all the way to 224×224 .
- **Middle-Frame Focus:** While we generate masks for the entire sequence, we prioritize the middle frame ($N//2$) as our primary output to ensure we are capturing the peak of the temporal information."

[3. Explainability: Transparency in AI] "Finally, we've built in **Explainability** features that go beyond standard detection:

- **Confidence Maps:** We calculate exactly how confident the model is about each pixel, where values closer to **0** or **1** represent higher certainty.
- **Temporal Stability:** We measure how much the prediction changes from frame to frame. A stable mask across time suggests a more reliable detection of a consistent forgery.
- **Domain Attribution:** We can even trace the detection back to its source—telling us whether the spatial, frequency, or noise branch was the primary driver for the decision."

[Conclusion] "By combining high-level classification with pixel-perfect localization and explainability, our model provides a transparent and robust defense against modern deepfake technology."

The training strategy for **AFAGNet v3** is designed to handle real-world challenges like class imbalance and missing data while ensuring the model remains stable during optimization.

1. The Multi-Task Loss Function

The total loss ($\mathcal{L}_{\text{total}}$) is a weighted combination of four specialized objectives:

- **Classification Loss ($\mathcal{L}_{\text{cls}}$):** We use **Focal Loss** instead of standard Binary Cross-Entropy (BCE). This is critical because our dataset has a **1:4.6 imbalance** (mostly fakes). Focal loss prevents the model from simply guessing "fake" every time and forces it to focus on "hard" samples—high-quality deepfakes that are difficult to distinguish.
- **Localization Loss ($\mathcal{L}_{\text{loc}}$):** To learn where the forgery is, we use a mix of ground-truth masks and **pseudo-masks**. Pseudo-masks are generated from the Frequency and Noise branches, allowing the model to learn localization even for datasets like FaceShifter that don't provide ground-truth masks.
- **Consistency Losses:**
 - **Branch Consistency ($\mathcal{L}_{\text{cons}}$):** Encourages the Spatial, Frequency, and Noise branches to semantically align.
 - **Temporal Consistency ($\mathcal{L}_{\text{temp}}$):** Uses a **clamped** difference between consecutive frame masks to prevent unstable frames from causing "exploding gradients" during training.

2. Scheduled Supervision

We use a **scheduled decay** for the localization weight (λ_{loc}).

- **Early Training:** We use high spatial supervision (0.8) to help the model find forgery regions.
- **Late Training:** We decay this weight down to 0.1 . This prevents the localization task from "overpowering" the classification task, ensuring the final "Real vs. Fake" verdict remains accurate.

3. Optimization & Stability

To ensure the model converges effectively on hardware like the RTX 3050, we use **AdamW** with a specialized learning rate strategy:

- **Differential Learning Rates:** The pretrained MobileViT-v2 backbone is trained at **10%** of the base learning rate to preserve its existing features, while the new modules (like CGAF) train at the full rate.

- **Warmup & Cosine Annealing:** We start with a 2-epoch linear warmup to stabilize the initial gradients, followed by a cosine decay to fine-tune the weights.
- **Exponential Moving Average (EMA):** We maintain a "smoothed" version of the model weights (0.9999 decay). This typically provides a **0.5–1.5% AUC improvement** by averaging out noise in the training trajectory.

"Our training isn't just about minimizing error; it's about balance. By using Focal Loss for class imbalance, pseudo-masks for missing data, and EMA for stability, we ensure AFAGNet v3 is as robust in training as it is in inference."